Akshat Patil

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Education

- **Medhavi Skills University** Sep 2025 – Jun 2029
  B.Tech in Computer Science & Engineering (AI/ML) *Bengaluru, KA*

Experience

- **Processing Foundation** Dec 2025 – Present
  Open Source Contributor & WebGPU Graphics Steward
  - Selected as an OSS Micrograntee '26 to design and implement a beginner-friendly WebGPU instancing API for p5.strands, later merged into p5.js core
  - Engineered and merged WebGPU features including primitive instancing, GPU-to-CPU storage-buffer reads, single-element buffer updates and automatic compute-dispatch handling
  - Serve as a WebGPU graphics steward for p5.js, reviewing contributions and shaping the direction of WebGPU development
  - Contributed 35+ merged PRs across p5.js core, Web Editor and website, including WebGL visual regression testing and shader/API improvements
- **Open Source Contributor & Maintainer** | *KubeStellar · Hiero · RISC-V · Paro Studio* 2026 – Present
  - Contributed 85+ merged PRs across KubeStellar, Hiero, RISC-V Unified DB, GreedyBear, NanoCoder and other open-source projects spanning Kubernetes, SDKs, developer tooling and infrastructure
  - Ranked #3 contributor in CNCF KubeStellar, filing 500+ verified bug reports covering i18n gaps, silent failures and broken state handling, alongside fixes across its console and monitoring ecosystem
  - Maintain Paro Studio, an open-source AI prompt gallery built with React, TypeScript and Supabase, managing the repository and coordinating 10+ contributors, with CI running lint, typecheck, tests and build on every PR

Projects

- **lookout** | *TypeScript · Express · PostgreSQL · Redis · React* GitHub | Live
  - Built a URL uptime monitoring platform with an Express API and a separate BullMQ worker on Redis, scheduling per-URL checks and classifying failures into 7 error types (DNS, TLS, timeout, refused and more)
  - Cut the dashboard's most frequent query from 50 ms to 0.5 ms by replacing `DISTINCT ON` with a `LATERAL` join on a composite index, and kept 7d/30d views fast with hourly rollups and raw-data retention
  - Added JWT auth with user/admin roles, structured logs with request IDs traced across API and worker, Docker setup and GitHub Actions CI, deployed on Vercel and Railway
- **Split-it-Wise** | *Next.js · Firebase · Firestore* GitHub | Live
  - Built and deployed a group expense splitter used daily by ∼20 hostel residents for real-money cost tracking
  - Handled all money maths in integer paise with unit tests, so balances always sum to zero with no float rounding drift, and enforced every permission server-side in Firestore rules rather than client code
- **DocRAG** | *Python · ChromaDB · Sentence Transformers* GitHub
  - Built a RAG-based QA system for technical PDFs using hybrid retrieval, combining MiniLM vector search with BM25 to handle both semantic queries and exact technical identifiers
  - Implemented PDF parsing, sliding-window chunking, score normalization and weighted retrieval fusion, with Gemini-generated answers grounded in retrieved chunks and page-level citations

Publications

- **Processing Foundation**: "Drawing a Forest in One Line: A Preview of Instancing in p5.strands", Aug 2026

Achievements

- **Processing Foundation OSS Micrograntee '26**: Paid grant to implement the WebGPU instancing API for p5.strands, nominated by project lead Kit Kuksenok from ∼120 GSoC applicants
- **CNCF KubeStellar – Admiral Badge**: Awarded for high-volume, high-impact contributions to KubeStellar
- **Mintlify Open Source Program '26**: Paro Studio selected for the 2026 open-source program

Technical Skills

- **Languages**: TypeScript, JavaScript, Python, Go, SQL
  Frontend: React, Next.js, Vite, Tailwind CSS, shadcn/ui
  Backend: Node.js, Express.js, REST APIs, Prisma, BullMQ, JWT Auth
  Databases: PostgreSQL, Redis, Supabase, Firebase/Firestore, MongoDB, MySQL
  DevOps & Tools: Docker, Kubernetes, GitHub Actions, Vercel, Railway, Linux, Git
  AI/ML: RAG, Embeddings, ChromaDB, scikit-learn, Pandas, LLM APIs (Gemini)
  Graphics/GPU: WebGL, WebGPU, GLSL/WGSL